

Joint Optimization of IRS-Aided MIMO VLC System Capacity Based on DRL

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Abstract—As an emerging technology, intelligent reflecting surface (IRS) can significantly improve the performance of visible light communication (VLC) by digitally adjusting the reflection parameters. In this paper, we consider improving VLC performance by deploying IRS in a multiple-inputs multiple-output (MIMO) environment. Firstly, we established the pairing relationship between IRS, light emitting diode (LED) and receiving photodiode (PD) and gave them constraints. Then we introduced the reflection coefficient vector of IRS and the power allocation vector of LED. Finally, through the deep reinforcement learning (DRL), we propose a twin delayed deep deterministic policy gradient (TD3) algorithm based on markov decision process (MDP) to solve the optimization problem of maximizing the capacity. Moreover, we have compared deep deterministic policy gradient (DDPG) algorithm to TD3. The last numerical results show the convergence, stability and effectiveness of the TD3 algorithm.

I. Introduction

With the rapid development of wireless communication technology, there is a growing demand for ultra-large capacity in communication systems. To satisfy this demand, the communication system not only widely adopts various technologies to improve the spectral efficiency of radio frequency (RF) communication, such as large-scale multiple-inputs multiple-output (MIMO) and non-orthogonal multiple access (NOMA), but also builds a new physical layer architecture which include visible light, millimeter wave (mmWave) and terahertz (THz) frequency as an improvement and supplementation for RF communication [1].

The pace of optical wireless communication (OWC) has been very fast that visible light communication (VLC) which based on using prevalent light emitting diode (LED) lighting devices has been a valuable OWC technology. While existing wireless technologies are depleting the electromagnetic (EM) spectrum available in RF bands, VLC utilizes unregulated, undeveloped, and large EM bands in the visible range, which means that VLC has the advantage of being free of spectrum licensing. VLC also has many favorable benefits, such as broad bandwidth, less interference with RF communications, excellent ability for reusing high frequency [2] and higher power efficiency [3]. Meanwhile, MIMO technology has been considered to achieve high data rates and reliability without consuming additional time and spectrum resources of VLC system [4]

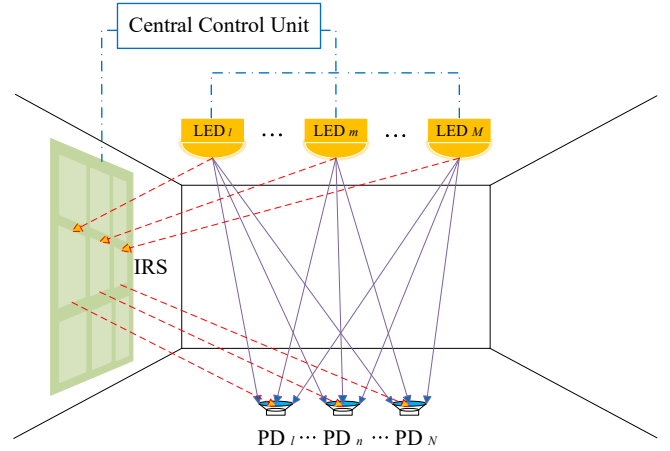


Fig. 1: Tunable IRS-Aided MIMO VLC System.

[5]. But visible light channels normally follow the typical Lambertian model, so that the channel gain depends on the position of the transceiver. When the receiving photodiode (PD) distance is close, the MIMO channels have a serious correlation and the channel matrix condition number is too large to obtain the ideal multiplexing gain [6]. To increase the channel capacity, the correlation of MIMO VLC channels is detailed by singular value decomposition (SVD) of the signal of 4×4 MIMO VLC system, and the system performance is improved by adaptive power and bits allocation strategy [7].

Since the VLC system mainly depends on the line of sight (LoS), and the signal gain of non-line of sight (NLoS) is negligible, the communication link will be interrupted and the situation will be affected once the LoS link is blocked. Nowadays, the emerging intelligent reflecting surface (IRS) start owning the capability of reconfiguring the wireless electromagnetic environment. Optical IRS overcomes the above shortcomings by providing NLoS optical signal transmission paths. Through controlling the relevant parameters of the IRS, the VLC system can surmount the high dependence of LoS, enhance the received signal power, expand the communication range and relax the dimming constraints on the basis of meeting

the lighting needs of users [6].

Recently, with the development of optical metasurface technology [8], the sensitivity and response speed of the IRS to visible light have been improved quickly and steadily [9] [10]. An innovative IRS element with adjustable phase and amplitude is proposed, which also has the function of reflected signal power control [11]. Deep-Q-learning is used to optimize resource management and maximize the spectral efficiency [12].

We aim at optimizing the pairing matrix for the IRS and maximizing the capacity of IRS-aided MIMO VCL system. And in this system, the reflection coefficient is adjustable and waiting to be optimized. In paper [13], it modeled IRS-reflected channel and characterize the IRS-aided MIMO VLC capacity under different transmitter emission power constraints. We will use the capacity of the system as a maximization function. The main contributions in our work are summarized as follows:

- First, we established a model of IRS-aided MIMO VCL system as shown in Fig.1. Based on the above, we constructed a general channel model between the signal transmitter and receiver, and it also follows the corresponding dimming power constraints and pairing relationships between the LED, IRS and PD.
- Next, we put forward an optimization issues which aims at maximize system capacity characterized in [13]. Because the parameters to be optimized include the reflection coefficient of the IRS, the LED power constraint, and the pairing relationship between the IRS, LED and PD, which is a non-convex mixed integer optimization problem, it is difficult to solve directly. Therefore, we established a markov decision process (MDP), and use twin delayed deep deterministic policy gradient (TD3) algorithm to find the optimal solution when the capacity is maximum.
- The final results show that the algorithm based on TD3 are effective and convergent, meaning that the calculational complexity is substantially decreased and more efficient and equitable resource allocation scheme can be solved for IRS-aided VLC system.

The rest of this article consists of the following. In Section II, a spatial system and channel model is constructed, and in Section III we proposed a TD3 algorithm based on deep reinforcement learning (DRL) to solve the optimal solution. Section IV provides numerical results to evaluate the performance of the proposed algorithm in IRS-aided MIMO VLC system. In the end, Section V summarizes the article.

II. MODEL AND PROBLEM FORMULATION

A. System Model

Consider an IRS-aided MIMO VLC system spatial model as shown in Fig. 1. There are M LEDs, N PDs, and K IRS elements in the system. By changing the reflection coefficient of the IRS element, the spatial freedom degree

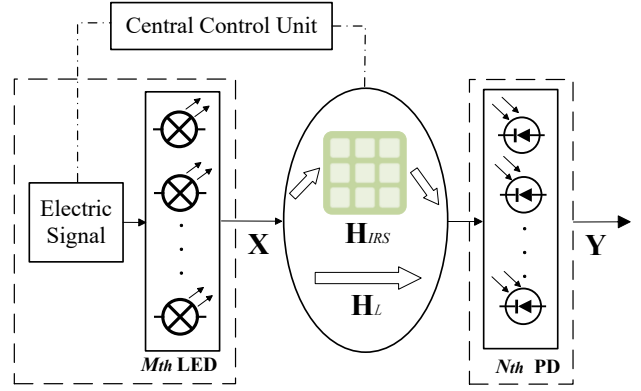


Fig. 2: IRS-Aided MIMO VLC System Model.

of the MIMO system can be improved, and the capacity can be further improved. That is, the IRS-aided MIMO VLC system can change the channel matrix and reduce the correlation of the MIMO channel through changing the IRS reflection.

B. Channel Model

Consider an $M \times N$ dimensional IRS-aided MIMO VLC system as shown in Fig. 2. The reflection coefficients of K IRS elements in the system are represented by the vector $\mathbf{r} = (r_1, \dots, r_K)$, $0 \leq r_k \leq \alpha$, and the r_k represents the maximum value of the reflection coefficient. When the k -th IRS element is operating, the central control unit controls the intensity of the reflected light from the IRS elements by setting the r_k .

The vector of the transmitted signal of M LEDs can be represented by $\mathbf{X} = (X_1, X_2, \dots, X_M)^T$. There are N PDs receiving signals at the same time. The received signal vector is $\mathbf{Y} = (Y_1, \dots, Y_N)^T$, and it has the following relationship with $\mathbf{X}$:

$$\mathbf{Y} = \mathbf{H}\mathbf{X} + \mathbf{Z} \quad (1)$$

where $\mathbf{Z} = (Z_1, \dots, Z_N)^T$ is an additive gaussian white noise vector, namely $\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_N)$.

In the presence of IRS, the channel matrix includes the direct path gain $\mathbf{H}_L$ of LEDs and the reflected path gain $\mathbf{H}_{IRS}$ of IRS elements. Thus, the channel matrix can be expressed as $\mathbf{H} = [\mathbf{H}_L + \mathbf{H}_{IRS}]$, $\mathbf{H} \in \mathbb{R}^{N \times M}$. The gain of direct path can be expressed as

$$\mathbf{H}_L = \left[h_{nm}^{(L)} \right]_{N \times M} \quad (2)$$

where $h_{nm}^{(L)}$ represents the LoS channel gains from the m -th LED to n -th PD, which can be calculated from the

$$h_{n,k,m}^{(\text{IRS})} = \begin{cases} \frac{(\kappa+1)A_{\text{PD}}\cos^\kappa(\phi_{m,k}^{\text{LR}})\cos(\psi_{k,n}^{\text{RD}})\cos(\theta_{k,n}^{\text{RD}})}{2\pi(d_{kn}+d_{km})^2(\vec{\mathbf{N}}_k^{\text{IRS}})^T\vec{\mathbf{E}}_{k,n}^{\text{RD}}}T_s(\psi_{k,n}^{\text{RD}})g(\psi_{k,n}^{\text{RD}}), & 0 < \psi_{k,n}^{\text{RD}} < \Psi_C \\ 0, & \psi_{k,n}^{\text{RD}} > \Psi_C \end{cases} \quad (5)$$

relative position between the PD and LED:

$$h_{nm}^{(\text{L})} = \begin{cases} \frac{(\kappa+1)A_{\text{PD}}}{2\pi d_{nm}^2}\cos^\kappa(\phi_{n,m})g(\psi_{n,m})T(\psi_{n,m})\cos(\psi_{n,m}), & 0 < \psi_{n,m} < \Psi_C \\ 0, & \psi_{n,m} > \Psi_C \end{cases} \quad (3)$$

where A_{PD} is the detection area of the receiving PD, $\kappa = -1/\log_2(\cos \Phi_{1/2})$ is the Lambertian radiation order of the LED, the $\Phi_{1/2}$ is the half-radiation angle of the light source, d_{nm} represents the distance between the LED and PD, $\phi_{n,m}$ is the radiation angle from the m -th LED to n -th PD, $\psi_{n,m}$ is the angle of incidence from m -th LED to n -th PD, Ψ_C is the half-field of view of the PD, $g(\psi_{n,m})$ is the gain of the optical filter. The $T(\psi_{n,m})$ is the gain of the optical concentrator, which can be calculated by the following formula:

$$T(\psi_{n,m}) = \begin{cases} \frac{n_r^2}{\sin^2 \Psi_C}, & 0 < \psi_{n,m} < \Psi_C \\ 0, & \psi_{n,m} > \Psi_C \end{cases} \quad (4)$$

where n_r denotes the refractive index of the optical concentrator.

Regardless of the effect of the reflection coefficient, the channel gain reflected by the k -th IRS element from the m -th LED to n -th PD is expressed in $h_{n,k,m}$ and can be calculated by the Eq.(5) at the top of this page. In Eq.(5), d_{kn} and d_{km} respectively represent the distance from the k -th IRS element to the n -th PD and the m -th LED light source, $\phi_{m,k}^{\text{LR}}$ represent the radiation angle from the m -th LED light source to k -th IRS element, the $\theta_{k,n}^{\text{RD}}$ represents the radiation angle from the k -th IRS element to n -th PD, $\psi_{k,n}^{\text{RD}}$ represents the angle of incidence from the k -th IRS element to n -th PD, $T_s(\psi_{k,n}^{\text{RD}})$ and $g(\psi_{k,n}^{\text{RD}})$ still represent the optical concentrator gain and optical filter gain of PD, $\vec{\mathbf{N}}_k^{\text{IRS}}$ represents the unit normal vector of the k -th reflector surface and $(\vec{\mathbf{N}}_k^{\text{IRS}})^T$ represents the rank of the vector, $\vec{\mathbf{E}}_{k,n}^{\text{RD}}$ represents the directional unit vector from the k -th IRS element to the n -th PD. $\mathbf{H}_{\text{IRS}}$ can be represented in terms of binary matrices $\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_M] \in \{0, 1\}^{K \times M}$ and $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N] \in \{0, 1\}^{K \times N}$, where $\mathbf{G}$ is the pairing between the IRS elements and the LEDs, $\mathbf{F}$ is the pairing between the IRS elements and the PD. Since the beam width of the reflected light of the IRS elements is highly spatially limited, there is no crosstalk between the IRS reflection channels and each IRS element can only provide

a reflection link for a pair of LED and PD. The binary matrix $\mathbf{G} \in \{0, 1\}^{K \times M}$ can be expressed as

$$\mathbf{G} = [\mathbf{g}_1, \dots, \mathbf{g}_M] = \begin{bmatrix} g_{1,1} & \cdots & g_{1,M} \\ \vdots & \ddots & \vdots \\ g_{K,1} & \cdots & g_{K,M} \end{bmatrix} = [\tilde{\mathbf{g}}_1, \dots, \tilde{\mathbf{g}}_K]^T \quad (6)$$

where $g_{k,m} = 1$ indicates that the k -th IRS element reflects the VLC signal emitted by m -th LED, and $\mathbf{g}_m$ and $\tilde{\mathbf{g}}_k$ respectively represent the m -th column and k -th row of $\mathbf{G}$. The binary matrix $\mathbf{F} \in \{0, 1\}^{K \times N}$ is used to represent the pairing between the IRS elements and PDs, which can be expressed as

$$\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_N] = \begin{bmatrix} f_{1,1} & \cdots & f_{1,N} \\ \vdots & \ddots & \vdots \\ f_{K,1} & \cdots & f_{K,N} \end{bmatrix} \quad (7)$$

where $f_{k,n} = 1$ means that the reflected light of the k -th IRS element is received by n -th PD, and $\mathbf{f}_n$ means that the n -th PD receives the VLC signal reflected by the IRS element. When $g_{k,m} = 1$ and $f_{k,n} = 1$, the k -th IRS elements reflects the signal from the m -th LED to n -th PD. When considering the reflection coefficient of the IRS elements, the channel gain $\mathbf{H}_{\text{IRS}} \in \mathbb{R}^{N \times M}$ through the reflection paths can be expressed as

$$\mathbf{H}_{\text{IRS}} = [(\mathbf{f}_n \odot \mathbf{g}_m)^T (\mathbf{r}^T \odot \mathbf{h}_{n,m})]_{N \times M} \quad (8)$$

where $\mathbf{h}_{n,m} = [h_{n,1,m}^{(\text{IRS})}, \dots, h_{n,K,m}^{(\text{IRS})}]^T$ represents the gain of the reflection path of K IRS elements between the m -th LED and the n -th PD without considering the reflection coefficient of IRS elements. $\odot$ denotes the Hadamard product. Therefore, we can define the channel matrix $\mathbf{H} = [\mathbf{H}_L + \mathbf{H}_{\text{IRS}}] = [h_{n,m}^{\text{total}}]_{N \times M}$, which the $h_{n,m}^{\text{total}}$ can be expressed as

$$h_{n,m}^{\text{total}} = h_{n,m}^{(\text{L})} + (\mathbf{f}_n \odot \mathbf{g}_m)^T (\mathbf{r}^T \odot \mathbf{h}_{n,m}) \quad (9)$$

Since each IRS element can only provide reflections for a pair of LED and PD, $g_{k,m}$ and $f_{k,n}$ need to meet the constraints:

$$\sum_{m=1}^M g_{k,m} \leq 1, \forall k \in \mathcal{K} \quad (10)$$

$$\sum_{n=1}^N f_{k,n} \leq 1, \forall k \in \mathcal{K} \quad (11)$$

In addition, the emitted light intensity of the LED needs to meet the peak power constraint and the dimming constraint:

$$0 \leq X_m \leq A_m, \quad m = 1, \dots, M \quad (12)$$

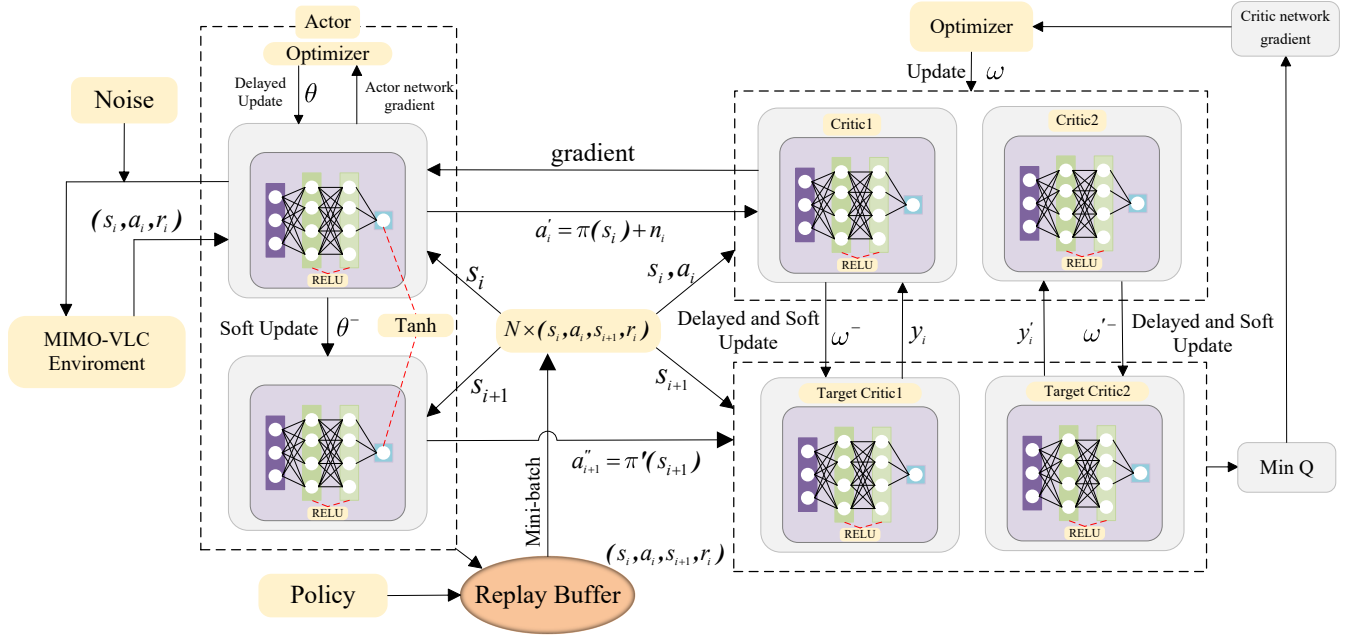


Fig. 3: TD3 principle

$$E[X_m] = E_m, \quad m = 1, \dots, M \quad (13)$$

where $\mathbf{A} = [A_1, \dots, A_M]^T$, $A_m > 0$ represents the maximum luminous intensity of M LEDs, and $\mathbf{E} = [E_1, \dots, E_M]^T$, $E_m > 0$ represents the average light intensity of the LED.

III. DRL-BASED ALGORITHM

A. Problem Formulation

In [12], it is stated that the progressive capacity of an IRS-aided MIMO VLC system can be given by the sum of the following common terms and a specific constant term:

$$f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}) = \frac{1}{2} \log \det [\mathbf{H}^T (\sigma^2 \mathbf{I}_N)^{-1} \mathbf{H}] + \sum_{m=1}^M \log A_m \quad (14)$$

where $\sigma^2 \mathbf{I}_N$ is the variance matrix of the additive gaussian white noise.

Therefore, in order to maximize the capacity of the IRS-aided MIMO VLC system, it is equivalent to maximizing $f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r})$. Based on the above situation, the capacity maximization problem studied in this paper can be expressed as follows:

$$\begin{aligned} P : \max_{\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}} \quad & f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r}) \\ \text{s.t.} \quad & f_{k,m}, g_{k,m} \in \{0, 1\}, \forall k \in \mathcal{K}, m \in \mathcal{M}, \\ & E_m > 0, \\ & 0 \leq r_k \leq \alpha, \\ & \sum_{m=1}^M g_{k,m} \leq 1, \forall k \in \mathcal{K}, \\ & \sum_{n=1}^N f_{k,n} \leq 1, \forall k \in \mathcal{K}, \\ & 0 \leq X_m \leq A_m, \quad m = 1, \dots, M, \\ & E[X_m] = E_m, \quad m = 1, \dots, M, \end{aligned} \quad (15)$$

It is important to note that Eq.(15) is difficult to obtain an optimal solution for the following reasons. First, the objective function $f(\mathbf{G}, \mathbf{F}, \mathbf{A}, \mathbf{r})$ is non-concave. Second, due to the binary constraints in $f_{k,m}$ and $g_{k,m}$, integer programming is also involved. Therefore, we proposed that the transmission parameter configuration scheme to maximize the capacity can be solved by DRL. The MDP state is the position and channel estimation of the receiver, the action includes the possible value of the transmission parameters, the policy is the transmission parameter configuration scheme that satisfies the constraints, and the reward function is the capacity.

IV. DRL-BASED ALGORITHM

In order to formulate the optimal strategy using the DRL-based algorithm, we construct an MDP so that the TD3 algorithm can be applied to solve Eq.(15). It is assumed that the IRS-aided MIMO VLC system is the environment and the central control unit is the agent.

A. MDP Model

In this MDP model, the state space, action space, reward, and state-action value function can be represented as follows:

- State space: We formulate the state space vector s_i as

$$s_i = [\text{vec}(\mathbf{G}), \text{vec}(\mathbf{F}), \text{vec}(\mathbf{A}), \text{vec}(\mathbf{r}), \text{vec}(\mathbf{H})] \quad (16)$$

where $\text{vec}(\cdot)$ is a vectorization operator. The dimension of the state space is $D_{state} = (M + N)K + M + K + NM$.

- Action space: Given that our goal is to obtain the optimal IRS correlation matrix, reflection coefficient vector and power allocation vector, we formulate the action vector $\mathbf{a}_i$ as:

$$\mathbf{a}_i = [\text{vec}(\Delta \mathbf{G}), \text{vec}(\Delta \mathbf{F}), \text{vec}(\Delta A), \text{vec}(\Delta \mathbf{r})] \quad (17)$$

where the dimension of the action space is $D_{action} = (M + N)K + M + K$. Noteworthily, the matrices $\mathbf{F}$ and $\mathbf{G}$ in the above formula are all binary, that means they need to be relaxed.

- Reward: Maximizing the capacity is the goal of the optimization problem, so the reward function r_t is defined as the capacity of the system in the current state:

$$\begin{aligned} r_i &= f(\mathbf{G}, \mathbf{F}, A, \mathbf{r}) \\ &= \frac{1}{2} \log \det [\mathbf{H}^T (\sigma^2 \mathbf{I}_N)^{-1} \mathbf{H}] + \sum_{m=1}^M \log A_m \end{aligned} \quad (18)$$

- State-action value function: Define $Q^\pi(\cdot)$ as state-action value function when the policy executes an action in the state. It can be expressed as follow on the basis of the Bellman equation,

$$Q^\pi(s_i, \mathbf{a}_i) = \mathbb{E}_\pi[r_i + \gamma Q^\pi(s_{i+1}, \mathbf{a}_{i+1}) | s_i, \mathbf{a}_i] \quad (19)$$

where $\gamma < 1$ is defined as the reward discount factor for the next state.

B. The TD3-Based Algorithm

The basic block diagram of TD3 is shown in Fig.3 and the process is shown in Algorithm 1. In the structure of the TD3 algorithm, there are six main networks which named actor network, critic1 network, critic2 network, target actor network, target critic1 network, target critic2 network. The actor network and the critic network are updated respectively by the backpropagation algorithm (BP) by parameters θ and ω , while the target actor network and two target critic networks are updated by θ^- , ω^- and ω'^- . We extract the current state from the environment and input it into the actor network to get the action, and input it into the critic1 network to generate a value function to evaluate the current action state function.

In the actor network, the dimension of the input layer is D_{state} , and the dimension of the output layer is D_{action} , and the Relu function is used as the activation function between layers. There are two hidden layers, and the number of neurons in each hidden layer is 880. Each layer is fully connected to each other, and the output of each layer is normalized by the Tanh function to improve learning performance and reduce the impact of initial values [14].

In two critic networks, the dimension of the input layer is $D_{action} + D_{state}$ and only one dimension in the output layer. This is because the critic network needs to input states and actions, and output a state-action

Algorithm 1: Joint Optimization based on TD3

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1: Initialize :
2: Actor network  $\mu_\theta(s)$  given  $\theta$ ;
3: Critic1 network  $Q_\omega(s, a)$  given  $\omega$ ;
4: Critic2 network  $Q_{\omega'}(s, a)$  given  $\omega'$ ;
5: Copy the same parameters  $\theta \rightarrow \theta^-$ ,  $\omega \rightarrow \omega^-$  and  $\omega' \rightarrow \omega'^-$  to initialize the target networks  $\mu_{\theta^-}$ ,  $Q_{\omega^-}$  and  $Q_{\omega'^-}$  respectively.
6: Set replaybuffer  $\mathcal{R}$  with size  $|\mathcal{B}|$  ;
7: Number of updates:  $n = 0$ 
8: for  $episode = 1 \rightarrow E$  do
9:   Initialize the Gaussian white noise  $\mathcal{N}$ ;
10:  Reset the initial state of the environment  $s_1$ ;
11:  for  $step = 1 \rightarrow T$  do
12:    Select action  $\mathbf{a}_t = \mu_\theta(s_t) + \mathcal{N}$  based on the current strategy and noise;
13:    Obtain  $(r_t, s_{t+1})$  from environment;
14:    Store the transition  $(s_t, \mathbf{a}_t, r_t, s_{t+1})$  into  $\mathcal{R}$ ;
15:    while  $|\mathcal{R}| > N$  do
16:      Sample  $N$  tuples  $e_i = \{(s_i, \mathbf{a}_i, r_i, s_{i+1})\}_{i=1, \dots, D}$  from  $\mathcal{R}$ ;
17:      For each tuple, using the target network to compute by (20);
18:      Minimize target losses by (21);
19:      Update the Critic network;
20:      Increase the number of updates:  $n = n + 1$ ;
21:      if  $n \% 3 == 0$  then
22:        Calculate the sampling policy gradient (22);
23:        Delayed update the Actor network;
24:        Delayed and soft update the target networks by:
25:           $\omega^- \leftarrow \tau \omega + (1 - \tau) \omega^-$ ;
26:           $\omega'^- \leftarrow \tau \omega + (1 - \tau) \omega'^-$ ;
27:           $\theta^- \leftarrow \tau \theta + (1 - \tau) \theta^-$ ;
28:        end if
29:      end while
30:    end for
31:  end for
32: return  $\mathbf{G}^*, \mathbf{F}^*, A^*, \mathbf{r}^*$ 

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value to evaluate the current policy. There are two hidden layers, and the number of neurons in each hidden layer is 880. The activation functions, output normalization, and connections between layers are consistent with the actor network. It is worth noting that in TD3, we need to take the minimum result of the two target critic networks, and then calculate the loss and gradient to update the neural network through the optimizer. The Adma optimizer is used for actor network and two critic networks to update.

During the training and updating of neural networks, what we do first is to store the sample to replay buffer. After this, we draw N samples from the replay buffer and calculate the target value function:

$$y_i = r_i + \gamma Q_{\omega^-}(s_{i+1}, \mu_{\theta^-}(s'_i)) \quad (20)$$

Next, we take the minimum value of the two target critic networks to calculate the loss function and immediately update two critic networks. The value of loss can be calculated as

$$L = \frac{1}{N} \sum_{i=1}^N \left[(y_i - Q_{\omega}(s_i, a_i))^2 + (y'_i - Q_{\omega'}(s_i, a_i))^2 \right] \quad (21)$$

And the update of the actor network parameters is determined by

$$\nabla_{\theta} J \approx \frac{1}{N} \sum_{i=1}^N \nabla_{\theta} \mu_{\theta}(s_i) \nabla_a Q_{\omega}(s_i, a) |_{a=\mu_{\theta}(s_i)} \quad (22)$$

Finally, the actor network, the target actor network and two target critic networks are delayed and softly updated. Particularly, the **F** and **G** matrices need to satisfy binary, but since the effect of noise on the action cannot make them absolutely binary, we binarize the **F** and **G** matrices after adding noise so that their values are absolutely 1 or 0.

V. SIMULATION AND RESULTS

A. Simulation scenarios

We set up a room of size $3m \times 3m \times 3.5m$ as shown in Fig.1 to simulate the environment. At the top of the room, we placed four LEDs in the middle as signal transmitters arranged evenly. Thirty six IRS elements are divided into four groups evenly distributed on the wall. Five PDs is placed evenly at the bottom of the room. The rest of the simulation parameters are shown as below.

TABLE I: SIMULATION PARAMETERS

Parameter	Description	Value
γ	Discount reward factor	0.985
$ \mathcal{B} $	The size of replay buffer	10^5
D	Number of samples in the mini-batch	128
E	Number of episodes	500
T	Number of steps in each episode	100
τ	Soft update coefficient	1×10^{-4}
V_a	Learning rate for actor network	1×10^{-5}
V_c	Learning rate for critic network	3×10^{-4}
A_m	Maximum luminous intensity	4
κ	The Lambertian index	1
$g(\psi_{n,m})$	Optical filter gain	1
A_{PD}	The area of PD	0.001
$\Phi_{1/2}$	Semi-angle of the FoV	70°
n_r	The optical concentrator refractive index	1.5
r_k	The maximum reflection coefficient	0.9

B. Simulation Results

In Fig. 4, we compared the resulting capacity between adjustable and fixed IRS reflection coefficients for the same number of thirty-six IRS elements. It can be seen that the final capacity of the adjustable IRS reflection coefficients/system is significantly higher than that of the fixed IRS reflection coefficient system. But at the same time, due to the introduction of adjustable IRS coefficients, the computational complexity of the algorithm are also increase, meaning that it are also slow down the convergence speed.

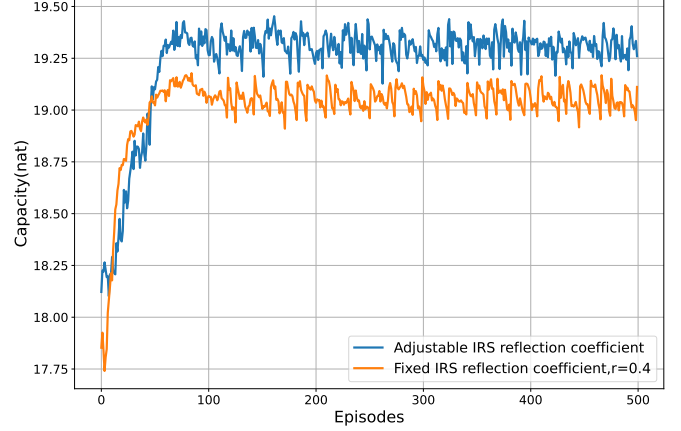


Fig. 4: Capacity with adjustable and fixed reflection

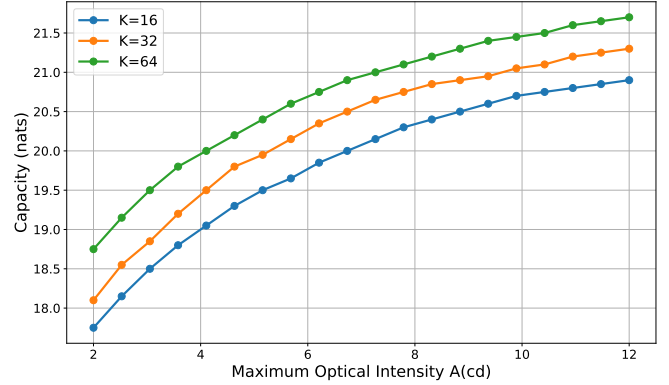


Fig. 5: Capacity with different maximum optical intensity.

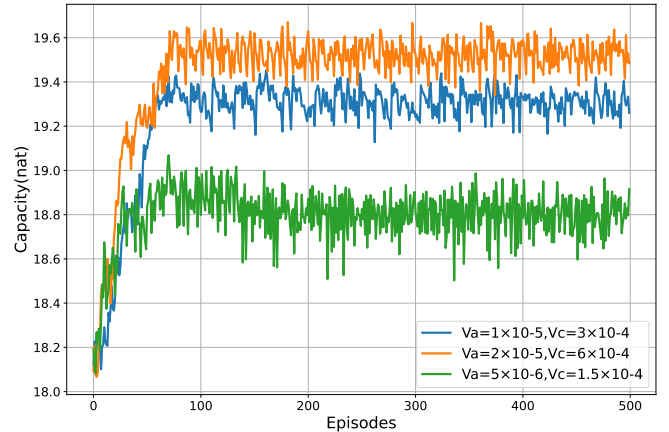


Fig. 6: Capacity with different learning rates TD3 .

Next, We compared the change in capacity with maximum optical intensity under different IRS in Fig. 5. When K is constant, the capacity of the system increases significantly as the maximum optical intensity increases.

We also compare the capacity at different learning rates when $K=36$ which can be seen from Fig. 6. As the learning rate changes, the maximum capacity that the system can achieve decreases, and the maximum capacity can only be reached when the actor network learning rate is equal to 1×10^{-5} . At the same time, as the learning rate decreases, the convergence speed slows down, and on the contrary, the system becomes unstable.

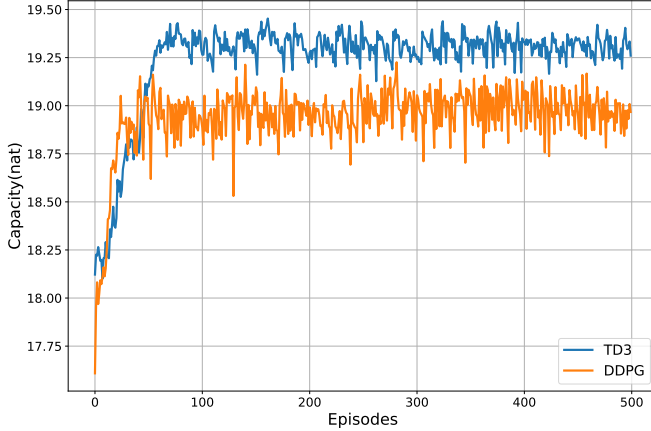


Fig. 7: Comparison of the capacity of DDPG and TD3.

Finally, in Fig. 7, we compared the different performance between the deep deterministic policy gradient (DDPG) algorithm and the TD3 algorithm in the same simulation environment. Since TD3 has an additional critic network than DDPG, the Q value of each critic network update is taken as the smaller of the two critic networks. Therefore, the TD3 algorithm can reduce the overestimation bias in DDPG and make the final algorithm results more stable and higher.

VI. CONCLUSIONS

In this paper, we jointly optimized the pairing matrix and reflection coefficient vector of IRS elements and the power allocation vector of LED based on the TD3 algorithm to maximize the capacity of the system. By describing the optimization problem as MDP, the computational complexity of the optimization problem can be significantly reduced. Finally, we also compare the performance of the respective algorithms of TD3 and DDPG. The results show that the TD3 algorithm successfully converges and calculates the final size of the maximum capacity, and also proves that TD3 has better performance than DDPG in maximizing capacity.

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